RENEWDESK PROTOTYPE NOTES

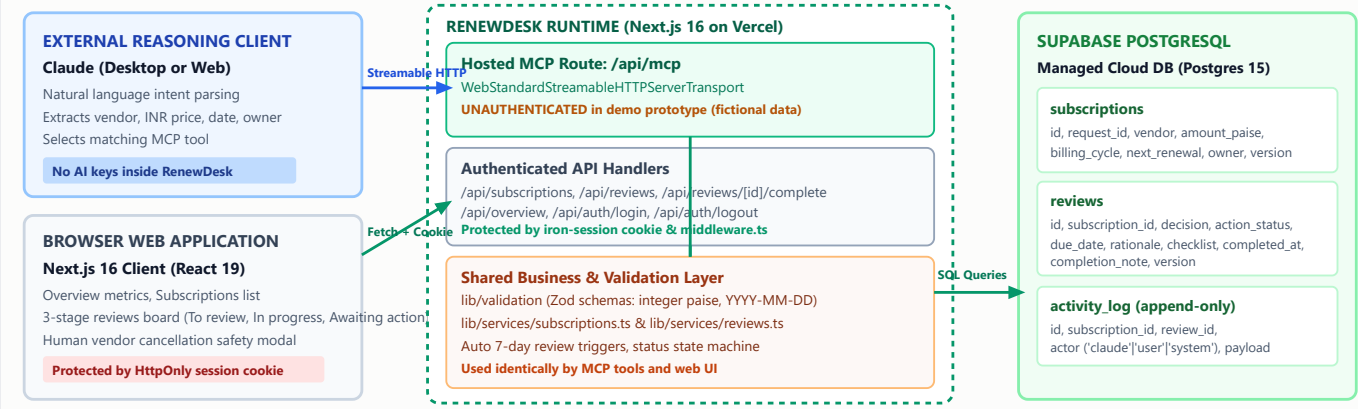
Architecture and Implementation Notes

Technical Briefing for Volopay Founder's Office, Product and Growth Role

Author: Sukhmeet Singh Lamba
Role Target: Founder's Office, Product & Growth
Live Application: `renewdesk-phi.vercel.app`
MCP Endpoint: `/api/mcp (Streamable HTTP)`

1. IMPLEMENTED SYSTEM ARCHITECTURE

Live in Production (32/32 Audit Checks Passing)



Clear Authentication Boundaries

Web UI: Protected by encrypted `iron-session` HttpOnly cookies. Middleware redirects unauthenticated visits to `/login` and blocks unauthenticated API routes with HTTP 401.

MCP Server: The endpoint at `/api/mcp` is unauthenticated for this prototype evaluation. It exposes fictional demo data only. Website session auth does NOT protect the MCP endpoint.

Reasoning vs. Storage Roles

Reasoning (Claude): Operates solely outside our application. Claude translates human conversation into structured tool invocations.

Persistence & Rules (RenewDesk): Validates payloads with Zod, ensures INR integer paise math, records 7-day triggers, and stores relational data in Supabase Postgres.

2. VERIFIED DATA FLOW: BIDIRECTIONAL CLAUDE & APP LIFECYCLE

Tested End-to-End

The complete cycle below was verified through our 32-check automated test suite and confirmed directly in Claude Desktop by Sukhmeet Singh Lamba:

1. **Claude Prompt:** User instructs Claude Desktop: "*Add a new subscription: Loom Pro, INR 2200/month, renews 2026-12-05, owned by Priya Sharma.*"
2. **MCP Tool Invocation:** Claude selects `save_subscriptions` and sends typed arguments over Streamable HTTP to `/api/mcp`.
3. **Validation & Auto-Review:** RenewDesk validates parameters with Zod, stores `amount_paise: 220000`, creates Loom Pro in PostgreSQL, automatically creates an active renewal review due 7 days prior in Asia/Kolkata (`2026-11-28`), and logs an entry to `activity_log` with actor `claude`.
4. **Web Dashboard Visibility & Human Edit:** User signs into RenewDesk. Loom Pro appears in 30-day renewal spend total, in Subscriptions, and in the "To Review" column on the Reviews board. User opens review and changes decision to `Cancel` with note: "*Migrating to Slack clips*".
5. **Human Safeguard & Read-Back:** Safety banner notifies user that setting Cancel does not stop billing; human confirmation checkbox is required. Claude queries `list_reviews` and cites Priya's rationale verbatim.

Retry Protection

Subscriptions accept an optional `request_id` parameter. If Claude retries a request, duplicate inserts are prevented via database constraint.

Optimistic Versioning

Both `subscriptions` and `reviews` carry integer `version` counters. Concurrent updates from web and Claude detect conflicts cleanly.

7-Day Review Triggers

Review creation is an on-write trigger during subscription insert. It is NOT an asynchronous background cron, keeping prototype execution deterministic.

Responsibilities, Tech Stack & Constraints

Author: Sukhmeet Singh Lamba
Prototype Context: Independent Assignment Submission

3. SYSTEM & PARTNER RESPONSIBILITIES

Actual vs. Proposed Boundaries

Explicit Prototype Boundary: RenewDesk is an independent assignment prototype created by Sukhmeet Singh Lamba. It has no implemented integration with Volopay systems, production card APIs, or banking rails. Volopay has no runtime role in this prototype.

Actor / System	Operational Boundary	Primary Responsibilities
Customer / Human Owner	Human in the loop	Provide subscription business context, assign decision owner, record keep or cancel verdict, and explicitly execute cancellation with vendor billing portal.
Claude / Anthropic	External AI Client	Reason over natural language inputs, extract structured parameters, select matching MCP tools, and present formatted query results to user. No app hosting.
RenewDesk Application	Core Software Boundary	Validate parameter schemas, enforce integer paise math, persist records, compute 30-day commit metrics, manage 3-column review stages, and enforce cancellation safety modal.
Supabase & Vercel	Managed Infrastructure	Vercel runs Next.js 16 serverless compute and Streamable HTTP endpoint. Supabase hosts managed PostgreSQL 15 database tables, triggers, and indices.
Subscription Vendor	External SaaS Provider	Execute recurring credit card charges, provide invoice receipts, enforce contract terms, and process account cancellation.
Volopay (Current Prototype)	No Runtime Role	Assignment evaluator. No implemented connection, live API usage, or financial data exchange exists in this deployed build.
Volopay (Proposed Partnership)	Future Financial Rail	Proposed corporate card integration: stream card transactions to detect new renewals, and auto-freeze virtual card when a human confirms cancellation in RenewDesk.

4. IMPLEMENTED TECHNOLOGY STACK & DEPLOYMENT INVENTORY

Production Verified

Core Runtime & Dependencies (from package.json)

- **Framework:** Next.js 16.4.6 (App Router, Turbopack, React 19)
- **Language & Build:** TypeScript 5, Node.js 20+
- **Styling & Components:** Tailwind CSS v4, Base UI (`@base-ui/react`), Lucide React, `tw-animate-css`
- **MCP Server SDK:** `@modelcontextprotocol/sdk` v1.32.2 (WebStandardStreamableHTTPServerTransport)
- **Database Client:** `@supabase/supabase-js` v2.119.0 (PostgreSQL 15)
- **Authentication:** `iron-session` v8.0.4 (Encrypted stateless cookie)
- **Validation:** `zod` v3.25.76 (Shared schemas for API and MCP)
- **Date & Timezone:** `date-fns` v4.4.0 & `date-fns-tz` v3.2.0 (Asia/Kolkata)
- **Notifications:** `sonner` v2.0.8 (Accessible toast alerts)

Live Deployment Coordinates & Configuration Names

- **Production App URL:** https://renewdesk-phi.vercel.app
- **Streamable HTTP MCP Endpoint:** https://renewdesk-phi.vercel.app/api/mcp
- **Demo Credentials:** `demo@renewdesk.app` / `RenewDesk2026!`
- **MCP Protocol Version:** `2024-11-05` (Supported tools: `list_subscriptions`, `save_subscriptions`, `list_reviews`, `save_reviews`)

Environment Variable Names (Values Kept Secret):
`SUPABASE_URL`, `SUPABASE_SERVICE_ROLE_KEY`, `SESSION_SECRET`, `DEMO_USERNAME`, `DEMO_PASSWORD`, `NEXT_PUBLIC_APP_URL`.

5. HONEST ASSUMPTIONS, CONSTRAINTS & LIMITATIONS

Documented Boundaries

Prototype Assumptions

- **Fictional Demo Data:** Seeded with 8 realistic SaaS subscriptions (Notion, Figma, Slack, GitHub, Linear, Google Workspace, AWS, Intercom).
- **Single Tenant Workspace:** Demo operates in one fixed organization (Acme Growth Studio). No multi-tenant team invites implemented.
- **User-Supplied Pricing:** Amounts and renewal dates come from user input or Claude interpretation. No automated bank scraping exists.

Honest Technical Limitations

- **Open MCP Endpoint:** The `/api/mcp` endpoint is unauthenticated for evaluator convenience. Production requires OAuth 2.0 or bearer keys.
- **Human Cancellation Required:** Setting a Cancel verdict never cancels the vendor contract. Human verification is strictly enforced in the UI.
- **Deterministic Triggers:** Review due date is calculated at write time (renewal date minus 7 days). It is not an asynchronous daemon scheduler.

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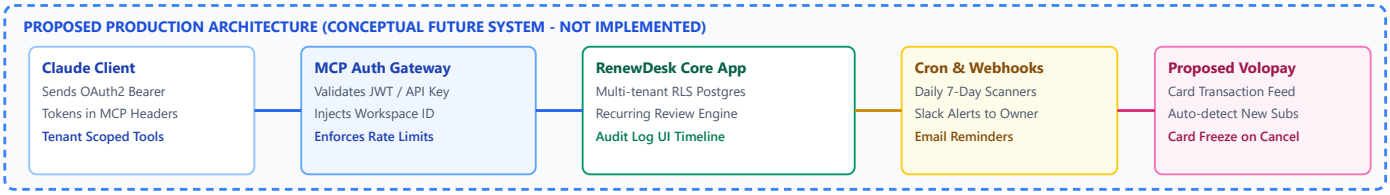
Built vs. Proposed System & Product Next Steps

Author: Sukhmeet Singh Lamba
Strategic Roadmap: From Prototype to Commercial Utility

6. BUILT PROTOTYPE VS. PROPOSED PRODUCTION ARCHITECTURE

Comparative Analysis

Architectural Area	What Exists Now (Prototype)	What Production Would Require	Why It Matters
Authentication & Workspaces	Single hardcoded demo user (`demo@renewdesk.app`) in one shared workspace.	Multi-tenant Supabase Auth, Row Level Security (RLS) policies by workspace ID, role-based access.	Prevents data leakage across different companies; allows team members to own reviews.
MCP Authorization	Public unauthenticated Streamable HTTP endpoint for evaluation ease.	OAuth 2.0 authorization code flow or user-scoped API bearer keys passed in headers.	Protects company financial commitments from unauthorized external scraping or injection.
Data Ingestion & Sync	Manual entry in web UI or conversational addition via Claude Desktop.	Email receipt forwarding parser and proposed Volopay virtual card transaction feed.	Eliminates manual entry; catches shadow IT renewals the team forgot to track.
Review Lifecycle	Single review per subscription, 7-day trigger calculated on write.	Recurring annual rollover: upon completion, automatically queues next year's review.	Prevents subscriptions from dropping out of tracking after their first reviewed renewal.
Alerts & Reminders	Visual badges (Overdue, Due in 7d) inside web dashboard only.	Scheduled Vercel Cron worker triggering Slack incoming webhooks and email digests.	Ensures founders and owners act before vendor cancellation notice deadlines lapse.
Observability & Audit	`activity_log` table written to Postgres on creates/updates; not surfaced in UI.	Surfaced audit timeline in subscription drawer and centralized OpenTelemetry metrics.	Provides transparency for executive spend audits and debugging AI tool calls.



7. PRIORITIZED ROADMAP: REDUCING REPETITIVE HUMAN WORK

Strategic Execution Plan

Priority 1: Authenticated Workspaces & OAuth MCP

Customer Problem: Multiple teams cannot safely use the app simultaneously without private workspace segregation.

Proposed Behavior: Implement Supabase Auth multi-tenancy and OAuth2 Bearer tokens for MCP tool discovery and execution.

Benefit: Enables production multi-company onboarding while strictly maintaining the no-AI application constraint.

Priority 2: Email-Based Renewal Discovery

Customer Problem: Subscriptions are forgotten until invoices arrive in an operator's inbox.

Proposed Behavior: Allow users to forward invoices to Claude or an ingestion inbox. Claude extracts renewal details via MCP.

Benefit: Cuts manual data entry to zero; software inventory stays synchronized automatically.

Priority 3: Slack Webhooks on 7-Day Triggers

Customer Problem: Operators forget to log into the web dashboard before cancellation notice windows close.

Proposed Behavior: Scheduled cron sends Slack DM to the assigned owner 7 days prior to renewal with direct link to decision card.

Benefit: Brings decision workflows directly into the team's existing daily communication channel.

Priority 4: Automated Multi-Cycle Annual Rollover

Customer Problem: After an annual subscription is reviewed, it drops out of the active review pipeline.

Proposed Behavior: Completing a review automatically advances `next_renewal` by 1 year and schedules the next review cycle.

Benefit: Creates sustained multi-year retention without requiring operators to manually recreate subscriptions.

Engineering & Product Sign-off: All live systems verified. Database, authentication, and Streamable HTTP MCP tools operating in production.

November 2026 | Sukhmeet Singh Lamba